

University of Pennsylvania

ESE5760 Final Project

Midterm Progress Report

Comparative Study of Memory Technologies and Process-Node
Scaling Trends Using DESTINY

Student: XXX
PennKey / ID: XXX
Instructor: XXX
Course: ESE5760
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1 Introduction

Emerging memory technologies such as eDRAM, STTMRAM, ReRAM, and PCRAM are increasingly important in architecture research because they offer different trade-offs in latency, dynamic energy, leakage power, and area. As memory capacity continues to grow, selecting the right technology becomes a multi-objective design problem rather than a single-metric optimization task.

This project uses DESTINY as the main modeling platform. DESTINY supports SRAM, eDRAM, and several non-volatile memory technologies in cache-style design exploration, making it suitable for studying cross-technology trade-offs under consistent assumptions. The long-term project goal includes three parts: fixed-capacity comparison across technologies, process-node scaling analysis, and 2D/3D structural comparison. At the midterm stage, the first two parts have been completed.

The purpose of this report is to summarize the completed work on:

1. fixed-capacity comparison across multiple memory technologies, and
2. process-node scaling trend analysis across several technology nodes.

2 Project Objectives

The current stage of the project focuses on the following objectives:

1. Compare SRAM, eDRAM, STTMRAM, ReRAM, and PCRAM under a fixed-capacity setup.
2. Evaluate their trade-offs in area, hit latency, hit dynamic energy, and leakage power.
3. Study how these trade-offs evolve as the process node scales from 90nm to 32nm.
4. Identify whether the relative ranking among memory technologies remains stable across process nodes.

3 Experimental Setup

3.1 Simulation Platform

All experiments in this report were conducted using the DESTINY memory modeling framework available in the project repository. DESTINY reports several key metrics relevant to architecture-level comparison, including:

- total area,
- cache hit latency,
- cache hit dynamic energy,
- total leakage power.

3.2 Fixed-Capacity Comparison Setup

To ensure a fair comparison across technologies, a common cache-style setup was used:

- Design target: cache
- Capacity: 2MB
- Associativity: 8-way
- Word width: 128-bit
- Process node: 45nm
- Routing: Non-H-tree
- Device roadmap: HP
- Optimization target: WriteEDP

The following five technologies were compared:

- SRAM
- eDRAM
- STTRAM
- ReRAM
- PCRAM

3.3 Process Scaling Setup

For the scaling study, the same architectural setup was preserved while only the process node was varied. The selected nodes were:

- 90nm

- 65nm
- 45nm
- 32nm

This setup allows the scaling trend to be isolated from other design variables.

4 Fixed-Capacity Comparison Across Memory Technologies

4.1 Experimental Results

Table 1 summarizes the fixed-capacity comparison across the five memory technologies.

Table 1: Fixed-capacity comparison across memory technologies

Technology	Area (mm ²)	Hit Latency (ns)	Hit Dynamic Energy (nJ)	Leakage Power (mW)
eDRAM	5.366	0.490	0.059	1010.998
PCRAM	0.803	0.663	7.730	355.567
ReRAM	3.668	1.462	0.159	267.058
SRAM	8.010	0.545	0.061	3587.418
STTRAM	4.933	2.887	0.131	258.759

4.2 Discussion

The results show a clear multi-objective trade-off among the five technologies.

From the latency perspective, SRAM and eDRAM provide the best performance. eDRAM achieves the lowest hit latency at 0.490ns, slightly better than SRAM at 0.545ns. This makes both technologies attractive for latency-sensitive cache designs.

From the area perspective, PCRAM is the densest option. Its area is only 0.803mm², which is far smaller than the other technologies. By contrast, SRAM occupies 8.010mm², the largest footprint among all candidates.

From the dynamic-energy perspective, PCRAM performs poorly. Its hit dynamic energy reaches 7.730nJ, which is dramatically higher than the others. SRAM and eDRAM remain near 0.06nJ, indicating that they are much more favorable in access-energy-sensitive scenarios.

From the leakage perspective, SRAM is the least favorable option, with 3587.418mW leakage power. eDRAM also has relatively high leakage at 1010.998mW. In contrast, STTRAM and ReRAM have much lower leakage, at 258.759mW and 267.058mW respectively.

4.3 Takeaway

The fixed-capacity study suggests the following:

- SRAM and eDRAM are strong latency-oriented choices.
- STTRAM and ReRAM are more attractive for leakage-sensitive designs.
- PCRAM is the most area-efficient option, but its dynamic-energy cost is extremely high.
- There is no universally best memory technology across all four metrics.

5 Process-Node Scaling Trend Analysis

5.1 Experimental Results

Table 2 reports the results across four process nodes for the five technologies.

Table 2: Process-node scaling results across memory technologies

Technology	Node (nm)	Area (mm ²)	Hit Latency (ns)	Hit Energy (nJ)	Leakage (mW)
eDRAM	32	2.685	0.442	0.035	1296.038
eDRAM	45	5.366	0.490	0.059	1010.998
eDRAM	65	10.494	0.811	0.099	2333.148
eDRAM	90	21.594	0.858	0.190	324.989
PCRAM	32	0.451	0.879	7.703	495.766
PCRAM	45	0.803	0.663	7.730	355.567
PCRAM	65	2.162	0.578	7.753	986.054
PCRAM	90	3.276	1.029	7.839	78.647
ReRAM	32	2.022	2.038	0.159	311.521
ReRAM	45	3.668	1.462	0.159	267.058
ReRAM	65	11.841	1.783	0.179	510.760
ReRAM	90	12.869	2.321	0.220	44.338
SRAM	32	4.059	0.595	0.036	4776.064

Technology	Node (nm)	Area (mm ²)	Hit Latency (ns)	Hit Energy (nJ)	Leakage (mW)
SRAM	45	8.010	0.545	0.061	3587.418
SRAM	65	16.744	0.704	0.110	5495.250
SRAM	90	32.025	0.918	0.197	1809.180
STTRAM	32	3.339	1.809	0.145	732.181
STTRAM	45	4.933	2.887	0.131	258.759
STTRAM	65	13.816	1.990	0.166	509.334
STTRAM	90	26.472	2.503	0.233	65.854

5.2 Overall Scaling Trends

A first-order observation is that area consistently decreases as the process node scales down. For example, SRAM shrinks from 32.025mm² at 90nm to 4.059mm² at 32nm, while eDRAM shrinks from 21.594mm² to 2.685mm². This confirms that technology scaling strongly improves storage density.

Dynamic energy also generally improves with scaling, especially for SRAM and eDRAM. SRAM drops from 0.197nJ at 90nm to 0.036nJ at 32nm, and eDRAM drops from 0.190nJ to 0.035nJ. However, this pattern is not equally strong for every technology. PCRAM remains near 7.7nJ across all nodes, suggesting that its access-energy behavior is dominated more by device assumptions than by CMOS scaling alone.

Latency generally improves as nodes scale, but the improvement is not always monotonic. SRAM and eDRAM remain the lowest-latency technologies across all nodes. STTRAM and ReRAM show some scaling benefit, but their latency remains clearly higher.

The most important observation is that leakage does not scale monotonically. For instance, eDRAM leakage rises from 1010.998mW at 45nm to 1296.038mW at 32nm, and SRAM leakage at 65nm is even higher than at 45nm and 90nm. This means that technology ranking is not preserved automatically as the process node changes.

5.3 Technology-Specific Interpretation

SRAM: SRAM consistently delivers low latency, making it the strongest performance-oriented baseline. However, its leakage remains very high across all nodes, and its area becomes especially costly at older nodes.

eDRAM: eDRAM provides latency and dynamic energy close to SRAM, while usually offering a smaller footprint. Its main drawback is that leakage behavior is irregular, so its efficiency cannot be assumed to improve monotonically with scaling.

STTRAM: STTRAM maintains a low-leakage advantage relative to SRAM, but its latency remains significantly worse. It is therefore better suited for leakage-sensitive rather than latency-critical designs.

ReRAM: ReRAM also offers relatively low leakage and moderate dynamic energy, but its latency remains noticeably slower than SRAM and eDRAM. It can be attractive when static power matters more than raw access speed.

PCRAM: PCRAM remains the most area-efficient design point across nodes. However, its dynamic energy stays extremely high and nearly node-insensitive, which limits its attractiveness for energy-conscious applications.

5.4 Takeaway

The scaling study leads to several conclusions:

- Process scaling improves area for all technologies.
- Dynamic energy usually improves, but not uniformly across technologies.
- SRAM keeps a latency advantage, while STTRAM and ReRAM keep a leakage advantage.
- PCRAM remains mainly an area-driven design point.
- Leakage is not monotonic, so memory-technology comparisons must be revisited at each node.

6 Current Progress

At the midterm stage, the project has completed the first two planned tasks.

First, a fixed-capacity comparison framework has been established, and quantitative results have been collected for five memory technologies across the key metrics of area, latency, dynamic energy, and leakage power.

Second, a process-node scaling study has been completed across four technology nodes. This study shows that process scaling changes not only absolute metrics but also the relative ranking among memory technologies, especially when leakage is considered.

These two completed parts already provide a solid basis for the final project, since they define the main cross-technology trade-off space and show how that space evolves with technology scaling.

7 Remaining Work and Next-Step Plan

Although the first two parts are complete, one major part of the project still remains: the comparison between 2D and 3D memory structures. The next phase will focus on:

1. comparing 2D, 2-layer, and 4-layer organizations,
2. analyzing the benefits and penalties of 3D stacking,
3. quantifying how area, latency, dynamic energy, and leakage change with stacking depth,
4. integrating all three parts into the final report.

In addition, the report can be further improved in the next phase by adding figures for clearer visual comparison, such as bar charts for fixed-capacity comparison and trend plots for process scaling.

8 Conclusion

This midterm report summarizes the completion of the first two main tasks of the project: fixed-capacity comparison across memory technologies and process-node scaling trend analysis.

The fixed-capacity results show that no single technology dominates across all metrics. SRAM and eDRAM are favorable for latency, STTRAM and ReRAM are favorable for leakage, and PCRAM is favorable for density but suffers from extremely high dynamic energy.

The scaling study further shows that technology ranking is not fixed across process nodes. While area generally improves and dynamic energy often decreases, leakage does not behave monotonically, which means the relative merit of each memory technology must be re-evaluated at each process node.

Overall, the completed work provides a strong quantitative foundation for the next stage of the project, which will extend the analysis to 2D/3D structure comparison and the benefits of memory stacking.